



Department of Computer Science

National University of Computer and Emerging Sciences
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Data Mining Assignment 1

Solution

Question 1: 3 marks

Compute the entropy for the random variable x

a. $P(x=\text{TRUE}) = 0.5$ $P(x=\text{False}) = 0.5$

b. $P(x=\text{TRUE}) = 1.$ $P(x=\text{False}) = 0$

c. $P(x=\text{TRUE}) = 0.2.$ $P(x=\text{False}) = 0.8$

Compute the entropy for the random variable x

a. $P(x=\text{TRUE}) = 0.5$ $P(x=\text{False}) = 0.5$ $H = -0.5 * \log(0.5) - 0.5 * \log(0.5) = 1$

b. $P(x=\text{TRUE}) = 1.$ $P(x=\text{False}) = 0$ $H = -1 * \log(1) - 0 * \log(0) = 0$

c. $P(x=\text{TRUE}) = 0.2.$ $P(x=\text{False}) = 0.8$ $H = -0.2 * \log(0.2) - 0.8 * \log(0.8) = 0.72$

Question 2: 18 points

Example	Pointed	Threaded	Width	Class
Example #1	No	Yes	Slim	Nail
Example #2	No	Yes	Slim	Nail
Example #3	No	No	Medium	Nail
Example #4	Yes	Yes	Fat	Nail
Example #5	Yes	Yes	Medium	Bolt
Example #6	No	Yes	Fat	Bolt
Example #7	No	Yes	Medium	Bolt
Example #8	No	No	Fat	Bolt

Let Class be the class label attribute.

- Calculate the Entropy of each feature. 3 points
- Calculate the Gini Index of each feature. 3 points
- Calculate the Information Gain of each feature and then report the best feature to split the examples using the information gain metric. 6 points
- Calculate the gain ratio for each feature and then report the best feature to split the examples using the gain ratio metric. 6 points

Best feature to split on -- Classification

1. (60 pts total) **Learning Decision Trees.** You are a robot in the fasteners section of a hardware store and must learn to discriminate Nails from Bolts. You choose to learn a Decision Tree classifier from various nails and bolts that you have observed. Unfortunately, your sensors are noisy, and so your training data has a few errors. This table summarizes the noisy, error-prone data you have accumulated for training:

$$H = -p \log(p) - (1-p) \log(1-p)$$

$$H(\text{Class}) = -4/8 * \log(4/8) - 4/8 * \log(4/8) = 1$$

Example	Pointed	Threaded	Width	Class
Example #1	No	Yes	Slim	Nail
Example #2	No	Yes	Slim	Nail
Example #3	No	No	Medium	Nail
Example #4	Yes	Yes	Fat	Nail
Example #5	Yes	Yes	Medium	Bolt
Example #6	No	Yes	Fat	Bolt
Example #7	No	Yes	Medium	Bolt
Example #8	No	No	Fat	Bolt

Pointed:

$$P(\text{Pointed}=Y) = 2/8 = 0.25$$

$$P(\text{Nail} | P=Y) = 1/2, \quad H(\text{class} | P=Y) = -1/2 * \log(1/2) - 1/2 * \log(1/2) = 1$$

$$P(\text{Nail} | P=N) = 3/6, \quad H(\text{class} | P=N) = -3/6 * \log(3/6) - 3/6 * \log(3/6) = 1$$

$$H(\text{Class} | \text{Pointed}) = 0.25 * 1 + 0.75 * 1 = 1$$

$$IG = H(\text{Class}) - H(\text{Class} | \text{Pointed}) = 1 - 1 = 0$$

Threaded:

$$P(T=Y) = 6/8 = 0.75$$

$$P(\text{Nail} | T=Y) = 3/6, \quad H(\text{class} | T=Y) = -3/6 * \log(3/6) - 3/6 * \log(3/6) = 1$$

$$P(\text{Nail} | T=N) = 1/2, \quad H(\text{class} | T=N) = -1/2 * \log(1/2) - 1/2 * \log(1/2) = 1$$

$$H(\text{class} | \text{Threaded}) = 0.75 * 1 + 0.25 * 1 = 1$$

$$IG = H(\text{Class}) - H(\text{Class} | \text{Threaded}) = 1 - 1 = 0$$

Width:

$$P(W=\text{Slim}) = 2/8, \quad P(W=\text{Medium})=3/8, \quad P(W=\text{Fat})=3/8$$

$$P(\text{Nail} | W=S) = 2/2, \quad H(\text{class} | W=S) = -1 * \log(1) - 0 * \log(0) = 0$$

$$P(\text{Nail} | W=M) = 1/3, \quad H(\text{class} | W=M) = -1/3 * \log(1/3) - 2/3 * \log(2/3) = 0.92$$

$$P(\text{Nail} | W=F) = 1/3, \quad H(\text{class} | W=F) = -1/3 * \log(1/3) - 2/3 * \log(2/3) = 0.92$$

$$H(\text{class} | \text{Width}) = 2/8 * 0 + 3/8 * 0.92 + 3/8 * 0.92 = 0.69$$

$$IG = H(\text{Class}) - H(\text{Class} | \text{Width}) = 1 - 0.69 = 0.31$$

which feature will be chosen to split on at root node?

- Calculate entropy of target
- Calculate IG of each feature
- Pick the feature with the largest IG (smallest $H(Y|X)$)

$$H(Y | X) = - \sum_{j=1}^c P(X = x_j) H(Y | X = x_j)$$

entropy of branch

prob of branch

$$H(Y | X) = - \sum_{j=1}^c P(X = x_j) \sum_{i=1}^k P(Y = y_i | X = x_j) \log_2 P(Y = y_i | X = x_j)$$

Question3: 9 points

- Find $P(A = t)$
- $P(B = f)$
- $P(C = t)$
- $P(B = t / C = t)$ (e) $P(A = f / C = t)$
- $P(A = t / C = f)$

(g) $P(A = f, C = t)$

(h) $P(A = t, C = t)$ (i) $P(A = t, b = f)$

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

4. (10 pts total, 1 pt each) Probability Rules and Independence.

Consider the following full joint distribution for Boolean variables A, B, and C:

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

$$P(A=t, B=t) = 0.03 + 0.12 = 0.15$$

$$P(A=t) = 0.03 + 0.12 + 0.17 + 0.18 = 0.5$$

$$P(B=t) = 0.03 + 0.12 + 0.03 + 0.12 = 0.3$$

$$P(A=t, B=t) = P(A=t) * P(B=t)$$

4.g. (1 pt) Are A and B independent of each other? (Y=Yes, N=No): Y

4.h. (1 pt) Are B and C independent of each other? (Y=Yes, N=No): N

<i>A</i>	<i>B</i>	<i>C</i>	<i>P(a,b,c)</i>
<i>t</i>	<i>t</i>	<i>t</i>	0.03
<i>t</i>	<i>t</i>	<i>f</i>	0.12
<i>t</i>	<i>f</i>	<i>t</i>	0.17
<i>t</i>	<i>f</i>	<i>f</i>	0.18
<i>f</i>	<i>t</i>	<i>t</i>	0.03
<i>f</i>	<i>t</i>	<i>f</i>	0.12
<i>f</i>	<i>f</i>	<i>t</i>	0.24
<i>f</i>	<i>f</i>	<i>f</i>	0.11

$$P(B=t, C=t) = 0.03 + 0.03 = 0.06$$

$$P(B=t) = 0.03 + 0.12 + 0.03 + 0.12 = 0.3$$

$$P(C=t) = 0.03 + 0.17 + 0.03 + 0.24 = 0.47$$

$$P(B=t, C=t) \neq P(B=t) * P(C=t)$$